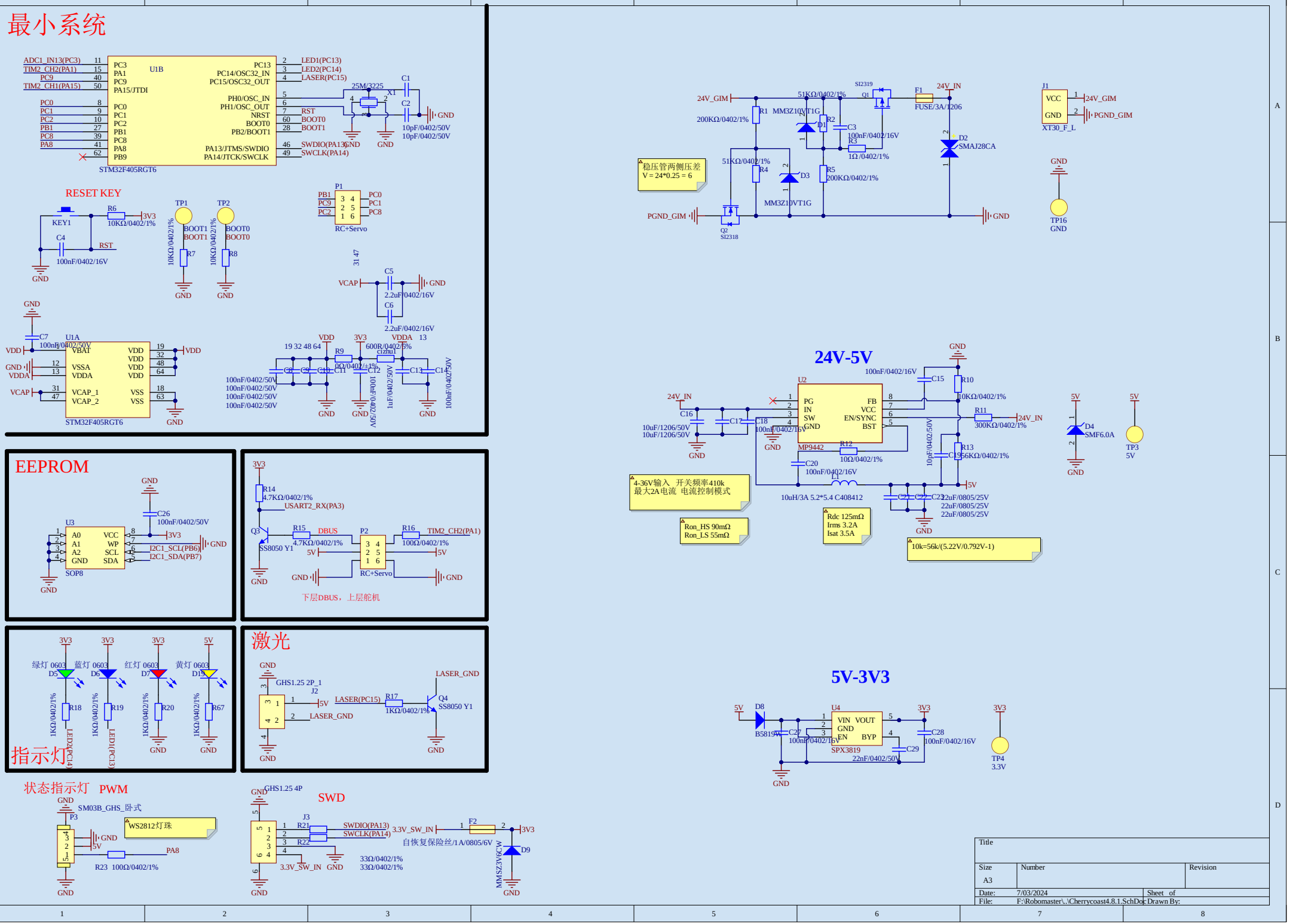
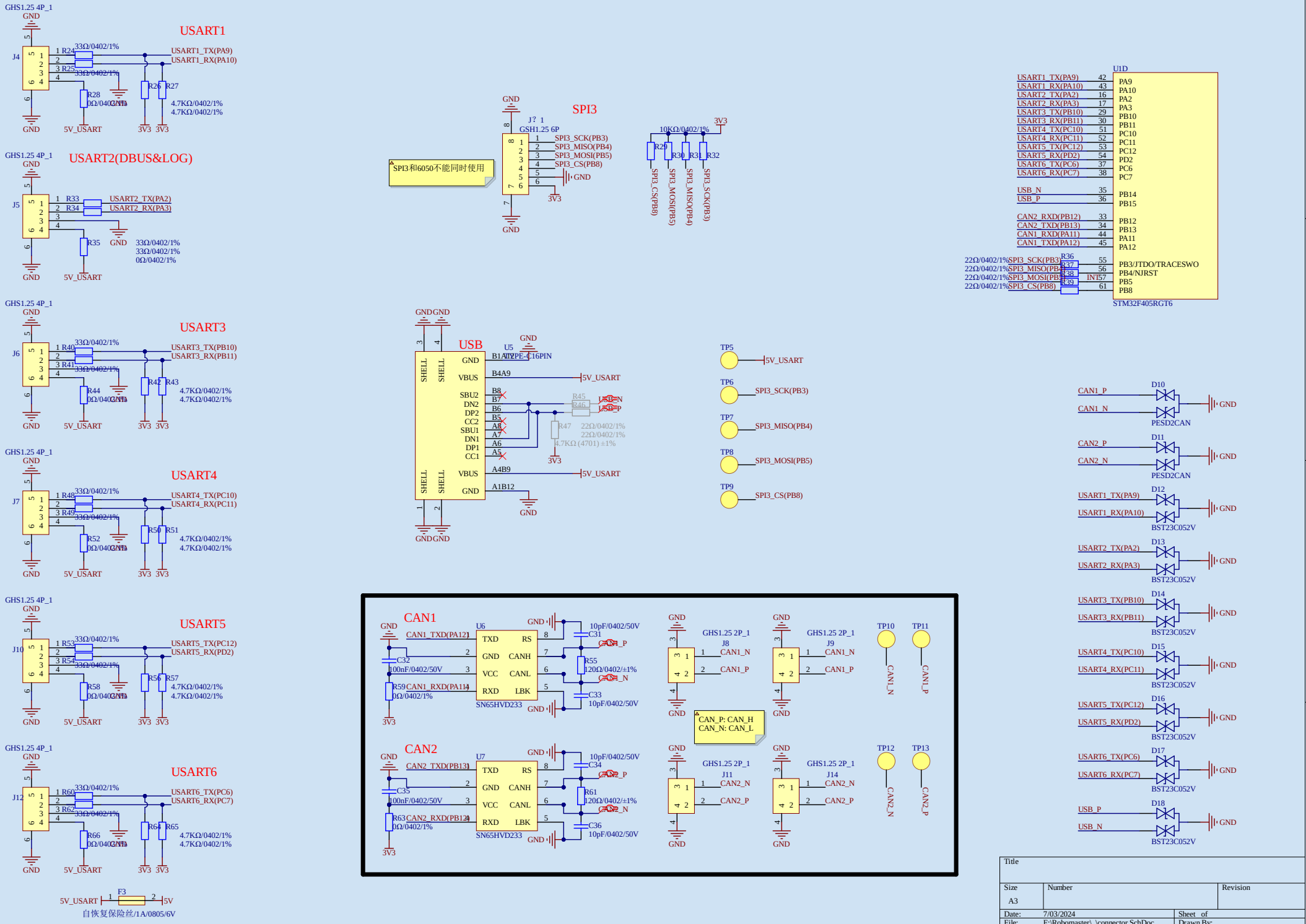
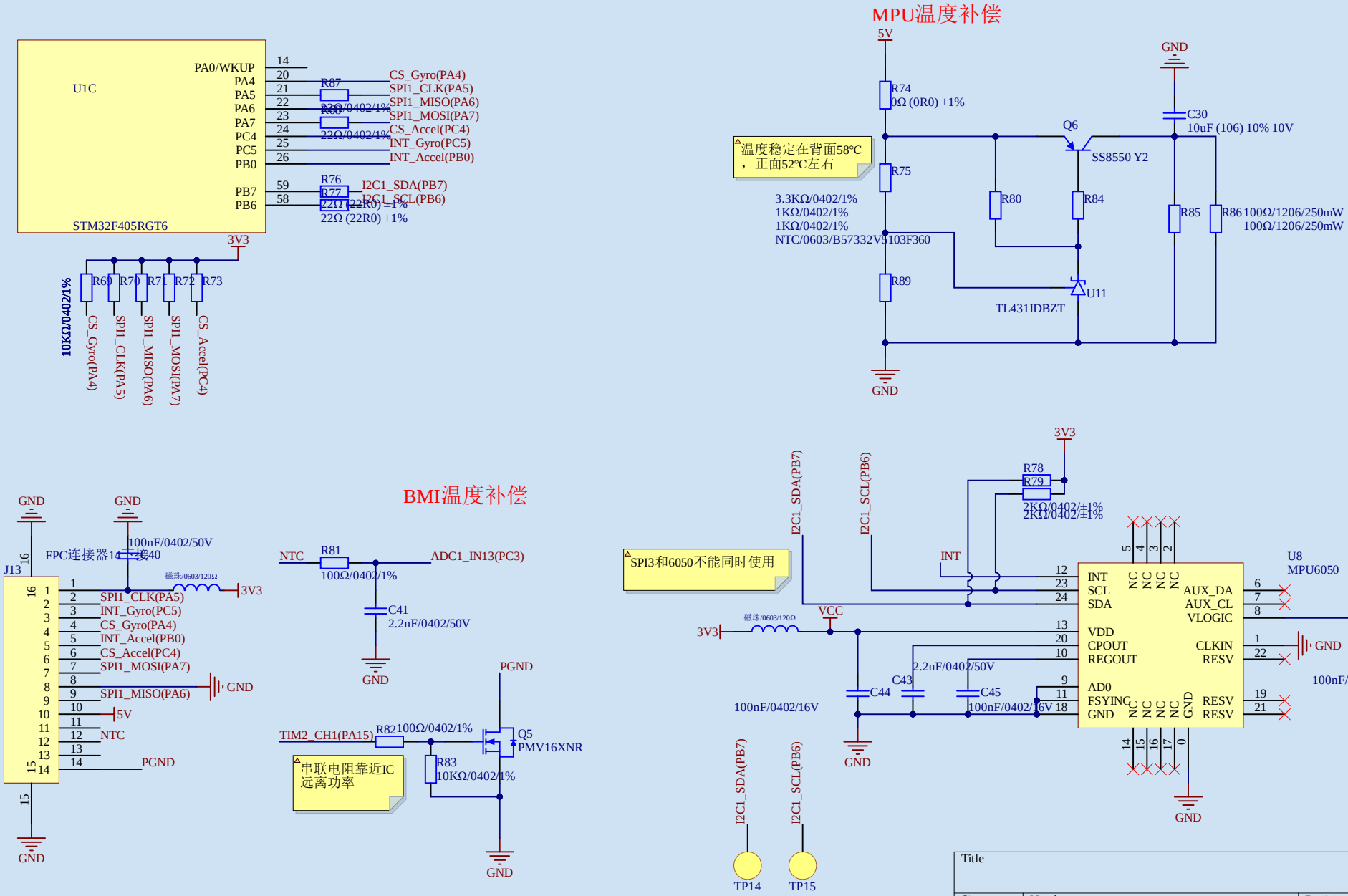

最小系统
STM32F405RGT6
RESET KEY
RC+Servo
EEPROM
SOP8
下层DBUS，上层舵机
指示灯
激光
状态指示灯 PWM
SWD
24V-5V
5V-3V3
稳压管两侧压差
V = 24*0.25 = 6
4-36V输入 开关频率410k
最大2A电流 电流控制模式
Ron_HS 90mΩ
Ron_LS 55mΩ
Rdc 125mΩ
Irms 3.2A
Isat 3.5A
10k=56k/(5.22V/0.792V-1)
WS2812灯珠
自恢复保险丝/1A/0805/6V
SPX3819
MP9442
Title
Size
A3
Number
Revision
Date: 7/03/2024
Sheet of
File: F:\Robomaster\..\Cherrycoast4.8.1.SchDoc
Drawn By:



Title			
Size A3	Number		Revision
Date: 7/03/2024		Sheet of	
File: F:\Robomaster\..\connector.SchDoc		Drawn By:	



Title		
Size A4	Number	Revision
Date: 7/03/2024		Sheet of
File: F:\Robomaster\..\IMU.SchDoc		Drawn By:

Comment	Description	Designator	Footprint	LibRef	Quantity
10pF/0402/50V	贴片电容	C1, C2, C19, C31, C33, C34, C36	0402_ding_c	0402 10pF (100) 5% 50V	7
100nF/0402/16V	贴片电容	C3, C4, C15, C18, C20, C27, C28	0402_ding_c	0402 100nF (104) 10% 16V	7
2.2uF/0402/16V	贴片电容	C5, C6	0402_ding_c	0603 2.2uF (225) 10% 16V	2
100nF/0402/50V	贴片电容	C7, C8, C9, C10, C11, C12, C14, C26, C32, C35, C40	0402_ding_c	0402 100nF (104) 10% 16V	11
1uF/0402/50V	贴片电容	C13	0402_ding_c	0603 1uF (105) 10% 50V	1
10uF/1206/50V	贴片电容	C16, C17	1206_ding_C	0603 100nF (104) 10% 50V	2
22uF/0805/25V	贴片电容	C21, C22, C23	0805_ding_C	0805 22uF (226) 20% 25V	3
22nF/0402/50V	贴片电容	C29	0402_ding_c	0402 22nF (223) 10% 50V	1
10uF (106) 10% 10V	贴片电容	C30	0603_ding_C	0603 10uF (106) 10% 10V	1
2.2nF/0402/50V	贴片电容	C41	0402_ding_c	0402 2.2nF (222) 10% 50V	1
100nF/0402/16V	贴片电容	C42, C44, C45	0402	0402 100nF (104) 10% 16V	3
2.2nF/0402/50V	贴片电容	C43	0402	0402 22nF (223) 10% 50V	1
600R/0402/5%	贴片电阻	cizhu1	0603_ding_R	0603 100Ω (1000) 1%	1
MM3Z10VT1G	稳压二极管 SOD-323 ±6%	D1, D3	SOD-323F	MM3Z10VT1G	2
SMAJ28CA	TVS	D2	SMAJ26CA	SMAJ26CA	1
SMF6.0A	稳压二极管	D4	SOD-123F_L2.8-W1.8-LS3.7-RD	LL-34 ZMM3V3	1
绿灯 0603	发光二极管	D5	0603_LED_S1	0603 绿灯	1
蓝灯 0603	发光二极管	D6	0603_LED_S1	0603 蓝灯	1
红灯 0603	发光二极管	D7	0603_LED_S1	0603 红灯	1
B5819W	肖特基二极管	D8	SOD-323F	SOD-123 B5819W SL	1
MMSZ3V6CW	肖特基二极管	D9	SOD-123	SOD-123 B5819W SL	1
PESD2CAN		D10, D11	SOT-23-3 L2.9-W1.3-P1.90-LS2.4-BR	PESD2CAN	2
BST23C052V		D12, D13, D14, D15, D16, D17, D18	SOT-23-3 TL431	BST23C052V	7
黄灯 0603	发光二极管	D19	0603_LED_S1	0603 黄灯	1
FUSE/3A/1206		F1	1206_R	FUSE/3A/1206	1
自恢复保险丝 /1A/0805/6V		F2, F3	0805_R	Reset fulse	2
XT30_F_L	卧式XT30	J1	XT30内嵌无极性标识_F	XT30_F_L	1
GHS1.25 2P_1	卧式	J2, J8, J9, J11, J14	SM02B-GHS卧式	GHS1.25 2P_1	5
GHS1.25 4P	立式	J3	SM04B-GHS立式	GHS1.25 4P	1
GHS1.25 4P_1	卧式短边	J4, J5, J6, J7, J10, J12	SM04B-GHS卧式	GHS1.25 4P_1	6
FPC连接器14下接		J13	FPC-SMD_14P-P0.50_XUNPU_FPC-05F-14PH15	FPC连接器14下接	1
GSH1.25 6P		J? 1	CONN-SMD_6P-P1.25_GH1.25-6PWBPZ	GSH1.25 6P	1
KEY	开关	KEY1	SW-SMD_L4.0-W3.0-LS4.8	KEY	1
10uH/3A 5.2*5.4 C408412	电感	L1	IND-SMD_L5.4-W5.2	L_47uH_T065125-磁环	1
RC+Servo	接口	P1, P2	RC+Servo	DBUS+Servo	2
SM03B_GHS_卧式		P3	SM03B_GHS卧式	SM03B_GHS_卧式	1
SI2319		Q1	SOT-23-3 TL431	SI2319	1
SI2318		Q2	SOT-23-3 TL431	SI2318	1
SS8050 Y1	三极管	Q3, Q4	SOT-23(SOT-23-3)	SOT-23 SS8050 Y1	2
PMV16XNR	MOS(场效应管)	Q5	SOT-23(SOT-23-3)	AO3400A N道沟 X0YV1T SOT-23	1
SS8550 Y2	三极管	Q6	SOT-23(SOT-23-3)	SOT-23 MMBT5401 2L	1
200KΩ/0402/1%	贴片电阻	R1, R5	0402_ding_R	0402 200KΩ (2003) 1%	2
51KΩ/0402/1%	贴片电阻	R2, R4	0402_ding_R	0402 51KΩ (5102) 1%	2
1Ω /0402/1%	贴片电阻	R3	0402_ding_R	0402 1Ω (1R00) 1%	1
10KΩ/0402/1%	贴片电阻	R6, R7, R8, R10, R29, R30, R31, R32, R69, R70, R71, R72, R73, R83	0402_ding_R	0402 10KΩ (1002) 1%	14
0Ω/0402/±1%	贴片电阻	R9	0402_ding_R	0402 0Ω (0R0) 1%	1
300KΩ/0402/1%	贴片电阻	R11	0402_ding_R	0402 300KΩ (3003) 1%	1
10Ω/0402/1%	贴片电阻	R12	0402_ding_R	0402 10Ω (10R0) 1%	1
56KΩ/0402/1%	贴片电阻	R13	0402_ding_R	0402 56KΩ (5602) 1%	1
4.7KΩ/0402/1%	贴片电阻	R14, R15, R26, R27, R42, R43, R50, R51, R56, R57, R64, R65	0402_ding_R	0402 4.7KΩ (4701) 1%	12
100Ω/0402/1%	贴片电阻	R16, R23, R81, R82	0402_ding_R	0402 100Ω (1000) 1%	4
1KΩ/0402/1%	贴片电阻	R17, R18, R19, R20, R67, R80, R84	0402_ding_R	0402 1KΩ (1001) 1%	7
33Ω/0402/1%	贴片电阻	R21, R22, R24, R25, R33, R34, R40, R41, R48, R49, R53, R54, R60, R62	0402_ding_R	0402 33Ω (33R0) 1%	14
0Ω/0402/1%	贴片电阻	R28, R35, R44, R52, R58, R59, R63, R66	0402_ding_R	0402 0Ω (0R0) 1%	8
22Ω/0402/1%	贴片电阻	R36, R37, R38, R39, R45, R46, R68, R87	0402_ding_R	0402 22Ω (22R0) 1%	8
4.7KΩ (4701) ±1%	贴片电阻	R47	0402_ding_R	0402 4.7KΩ (4701) 1%	1
120Ω/0402/±1%	贴片电阻	R55, R61	0402_ding_R	0402 120Ω (1200) 1%	2
0Ω (0R0) ±1%	贴片电阻	R74	0402_ding_R	0402 0Ω (0R0) 1%	1
3.3KΩ/0402/1%	贴片电阻	R75	0402_ding_R	0402 4.7KΩ (4701) 1%	1
22Ω (22R0) ±1%	贴片电阻	R76, R77	0402_ding_R	0603 22Ω (22R0) 1%	2
2KΩ/0402/±1%	贴片电阻	R78, R79	0402_ding_R	0402 2KΩ (2001) 1%	2
10Ω/1206/250mW	贴片电阻	R85, R86	1206_R	1206 22Ω (22R0) 1%	2
NTC/0603/B57332V5103F360	贴片电阻	R89	0603_R	0603 4.7KΩ (4701) 1%	1
BOOT1		TP1	TestPoint	TestPoint	1
BOOT0		TP2	TestPoint	TestPoint	1
5V		TP3	TestPoint	TestPoint	1
3.3V		TP4	TestPoint	TestPoint	1
		TP5, TP6, TP7, TP8, TP9, TP10, TP11, TP12, TP13, TP14, TP15	TestPoint	TestPoint	11
GND		TP16	TestPoint	TestPoint	1
STM32F405RGT6		U1	LQFP-64 10x10x05P	STM32F405RGT6	1
MP9442	No Description Available	U2	TSOT23-8_MP2161_MNP	MP9442	1
SOP8	24C02	U3	SOP8	AT24C02	1
SPX3819	LDO	U4	SOT-23-5	SPX3819	1
TYPE-C16PIN		U5	USB-C-SMD_TYPE-C16PIN	TYPE-C16PIN	1
SN65HVD233		U6, U7	SOIC-8 150mil	SN65HVD233	2
MPU6050		U8	QFN-24 mpu6050	MPU6050	1
磁珠/0603/120Ω		U9, U10	L0603	磁珠/0603	2
TL431IDBZT	Adjustable precision shunt regulator 3-SOT-23 -40 to 85	U11	SOT-23-3 TL431	TL431IDBZT	1
25M/3225	25M/3225	X1	SMD-3225 4P	XTAL 4	1